

Instituto Tecnológico de Costa Rica

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Curso:

Inteligencia Artificial

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JarvisTEC

A voice and vision assistant over ten machine learning models

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1 Proposed solution

1.1 Problem statement

The project asks for a personal assistant that hears the user, reads the user's face, and answers questions about a set of machine learning problems. Its scope has four deliverables: an agent that recognises the emotion of the person from a live camera frame, an agent that turns speech into text and executes the instruction, at least ten machine learning models chosen from the list of the course, and a set of spoken commands that run those models. The models are served by a REST API, and the user talks to them through a desktop application.

JarvisTEC is the answer to that statement. It is a Spanish-speaking desktop assistant in which every request follows one path: the microphone records the user, a speech-to-text model transcribes the audio, an intent parser decides which of the ten models the sentence refers to and which values it carries, the REST API runs that model, and the reply is written and spoken back in a tone that depends on the emotion the webcam read on the user's face. An animated penguin avatar changes its expression with that emotion.

1.2 Design decisions

Three decisions shape the solution and are defended here rather than in the code.

Everything runs locally. The statement allows cloud services from Google or Azure for face and speech recognition in the first delivery. JarvisTEC does not depend on them: the emotion classifier is a convolutional network exported to ONNX and run with `onnxruntime` on the CPU, speech-to-text is Whisper through `faster-whisper`, and text-to-speech is Piper with a Mexican Spanish voice. The only optional cloud call is to Gemini for intent parsing, and when no key is configured a rule-based parser takes over with the same contract, so the assistant works offline end to end. This makes the demo independent of quotas, network and billing, and keeps the user's camera frames and voice on the machine.

Ten models, one contract. Each model is a CRISP-DM notebook that ends by writing a `joblib` bundle with the estimator and everything the API needs to feed it without retraining: the predictor list, encodings, fill values for missing inputs, thresholds, class names and training ranges. The API loads the bundles once at startup and exposes them behind one endpoint shape, `POST /predict/{model_name}`. The application never touches an estimator; it only speaks JSON. Adding an eleventh model is a notebook, an adapter and a schema, and the assistant discovers the new schema at startup.

The emotion changes how the answer is said, not what it says. The classifier returns one of seven emotions (angry, disgust, fear, happy, sad, surprise, neutral), the vocabulary of the FER-2013 challenge, which in turn follows Ekman's basic emotions. The prediction of the model is computed first and is identical whatever the face shows; a tone table then wraps it in a prefix and a suffix (calm and short for an angry user, cheerful for a happy one). This keeps the machine learning answer auditable and confines the affective layer to phrasing.

1.3 Commands

The assistant wakes on the word *JarvisTEC* (and the mishearings Whisper produces for it, such as *Yarvis* or *Yalvis*, all matched on a word boundary). One sentence per model is enough when it

carries every value the model needs; when a required value is missing the assistant asks for it and keeps the pending intent across turns. Three example sentences per model, in Spanish, ship with the application in `app/prompts.txt`; Table 1 gives one per model.

Model	Example command (Spanish, as spoken)
Bitcoin	*JarvisTEC, ¿cuánto va a costar el bitcoin mañana?*
Car price	*JarvisTEC, ¿cuánto vale un carro diesel del 2015 que nuevo costó 10 lakhs, tiene 40 mil kilómetros, un solo dueño, lo vende una agencia y es manual?*
Churn	*JarvisTEC, ¿se va a ir un cliente hombre . . . contrato mes a mes, factura electrónica, paga con cheque electrónico, lleva 3 meses y paga 95 dólares al mes?*
Wine	*JarvisTEC, ¿qué calidad tiene un vino blanco con 13.5 de alcohol, densidad 0.99, . . . y 2 de azúcar residual?*
Flight delay	*JarvisTEC, ¿cuánto se va a retrasar el vuelo de American Airlines de Dallas a Chicago el lunes 15 de enero a las 12 del mediodía, con 802 millas?*
Stroke	*JarvisTEC, ¿qué riesgo de derrame tiene un hombre de 68 años, hipertenso, . . . con glucosa de 140, índice de masa corporal 31 y que dejó de fumar?*
Cirrhosis	*JarvisTEC, ¿tiene cirrosis una mujer de 55 años sin edema, sin ascitis, con hepatomegalia, . . . plaquetas 200 y protrombina 11?*
Avocado	*JarvisTEC, ¿a cuánto estaría el aguacate orgánico en San Francisco el 10 de septiembre de 2017 con un volumen total de 20 mil . . . ?*
Hepatitis C	*JarvisTEC, ¿qué categoría de hepatitis tiene un hombre de 45 años con albúmina 38, fosfatasa alcalina 60, ALT 30, AST 90, . . . y proteínas 70?*
Body fat	*JarvisTEC, ¿cuál es mi porcentaje de grasa corporal si mi abdomen mide 90 centímetros, peso 175 libras y mi muñeca mide 18 centímetros?*

Table 1: One spoken command per model. The full set, three per model, is in `app/prompts.txt`.

1.4 User interface

The window (Figure 1) has the avatar and the camera card on the left and the conversation on the right. The camera card shows the mirrored webcam feed with the detected face boxed and the emotion and its confidence written under it. The input row has a single button that is the microphone while the entry is empty, the send arrow once there is text, and a red recording indicator while the microphone is open. A status strip at the bottom reports the microphone, the emotion detector, the API, the speech-to-text engine and which parser answered (a Gemini model or the rules). System notices never enter the chat.

2 Machine learning architectures

2.1 System architecture

Figure 2 shows the path of one request. The application is split into three processes that share only files and HTTP: the notebooks produce the model bundles, the API serves them, and the desktop application consumes the API. Inside the application every stage is behind a protocol (an abstract interface), so the Whisper backend, the Piper backend, the ONNX emotion detector, the Gemini parser and the HTTP client are each replaceable without touching the orchestrator.

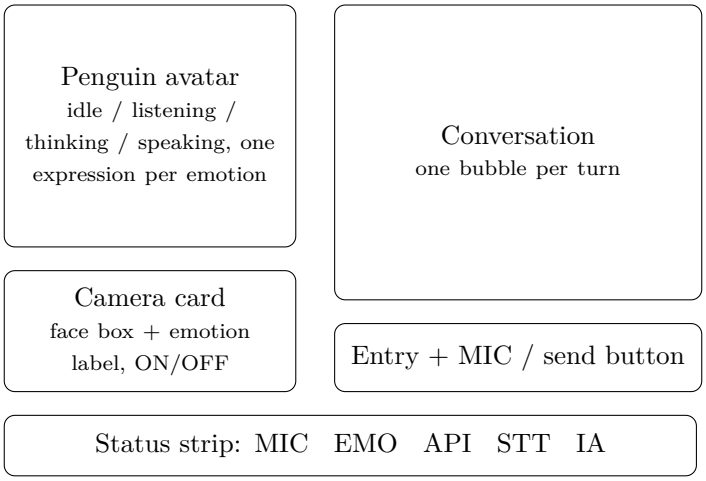


Figure 1: Layout of the JarvisTEC window. `Enter` sends typed text, `Ctrl+Space` toggles recording, `Ctrl+E` toggles the webcam, `Esc` quits.

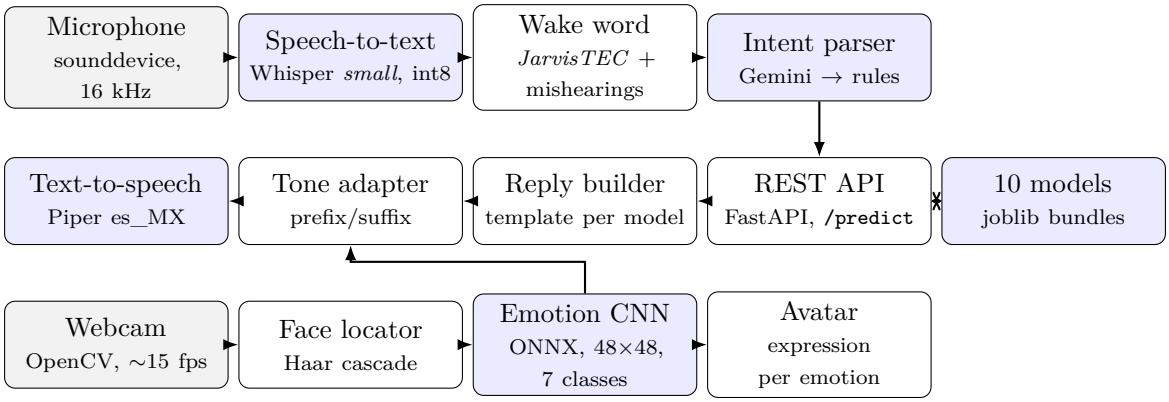


Figure 2: One request through JarvisTEC. Shaded boxes are learned models. Typed text enters at the wake-word stage and follows the same path.

2.2 Perception agents

Facial emotion agent. Every 2 seconds the camera worker takes the current frame, converts it to grayscale and runs the Viola–Jones Haar cascade (`haarcascade_frontalface_alt`) to locate the largest frontal face, rejecting boxes under 60×60 pixels to avoid glare and texture patches. The crop is resized to 48×48, scaled to $[0, 1]$ and fed to a VGG-style convolutional network: five blocks of 3×3 convolutions with ReLU (64, 128, 256, 512 and 512 filters, two or three convolutions per block), each block closed by batch normalisation and 2×2 max pooling, then a dense layer of 128 units with dropout 0.2 and a softmax over the seven classes. The network was trained on FER-2013 and reused from an earlier course project; for this project the Keras weights were exported once to ONNX so that inference runs with `onnxruntime` on the CPU with two threads and does not starve the interface. The winning class always drives the avatar and the tone; with seven classes a webcam reading rarely exceeds 50 % probability, so the confidence is shown for information only. The reading is repeated every two seconds and the face box is redrawn on every frame in between.

Speech agent. `Ctrl+Space` or the microphone button opens a 16 kHz mono recording; pressing again closes it. The audio goes to Whisper *small* through `faster-whisper` (CTranslate2), loaded in int8 on the CPU with Spanish fixed as the language. The transcript then follows exactly the same path as typed text: wake-word check, intent parsing, prediction, reply. Whisper's typical mishearings of the wake word are part of the wake-word table, and the Gemini prompt is told the text may come from speech so it can recover *días* from *dientes*.

Intent parser. The parser turns a Spanish sentence into `(model_name, fields, missing)`. With a key configured, Gemini is asked for a JSON object under a system instruction that carries the catalogue of the ten models (a one-line description, Spanish synonyms and per-model inference rules) and the JSON schema of every request body, which the application reads from the API's OpenAPI document at startup. Models are tried in a chain and a model whose quota is exhausted (429) or that was retired (404) is skipped for the rest of the session. Without a key, or on any Gemini failure, a rule-based parser with one field extractor per model (keyword lookup tables, Spanish number words, dates, airline and city codes) produces the same structure. Whichever parser answered is shown in the status strip.

Dialogue. The orchestrator records the turn, resolves the intent against the pending one (so an answer to "which airline?" completes the flight request from the previous turn), asks for the next missing required field if any, or posts the request body to the API and renders the answer through a per-model reply template. The tone adapter wraps the text according to the last emotion, and the speaking worker synthesises it with Piper (`es_MX-claude-high`, 22 kHz). Starting a new recording interrupts the voice.

2.3 The ten models

Table 2 lists the ten problems, their datasets, the technique required by the course for each and the target. Each one is developed in its own notebook with the six CRISP-DM phases: problem analysis, data understanding, exploration and preparation, modelling, evaluation and deployment (the saved bundle plus a reload test). Every notebook builds the literal model the task asks for first, and when the data call for a deviation (a logarithm, a regrouped target, a balanced class weight) that second model is built next to it and the choice is defended with the metrics of both.

#	Notebook	Dataset (source)	Technique	Target
1	`bitcoin_price_regression`	Bitcoin daily prices 2013–2017 (Kaggle, team-ai)	ARIMA	closing price, next days
2	`car_price_regression`	301 used cars, India (Kharwal)	multiple linear regression	selling price (lakhs)
3	`telecom_churn_classification`	IBM Telco churn, 7 043 customers	neural network (MLP)	churn yes/no
4	`wine_quality_classification`	Vinho verde, 6 497 wines	k nearest neighbours	quality band
5	`flight_delay_regression`	BTS airline delays 2008, 1.9 M flights	decision tree	arrival delay (min)
6	`stroke_classification`	Stroke prediction, 5 110 people (fedesoriano)	logistic regression	stroke yes/no
7	`cirrhosis_type_classification`	Mayo PBC trial, 418 patients	support vector machine	cirrhosis stage
8	`avocado_price_regression`	Hass Avocado Board retail scans 2015–2018	random forest	average price (USD)
9	`hepatitis_type_classification`	HCV blood panels, 615 people	Gaussian naive Bayes	donor / hepatitis / fibrosis / cirrhosis
10	`body_fat_regression`	252 men	multiple linear regression	body fat (%)

Table 2: The ten models, in the order of the repository. All datasets are downloaded by the notebooks through `kagglehub` or a public URL.

2.4 Serving architecture

The API is a FastAPI service with one adapter per model. Each adapter knows how to turn a validated request body into the row the estimator expects (filling optional fields with the notebook's fill value, applying the notebook's encodings and transformations) and how to turn the estimator's output into the response (label, probability and threshold for classifiers; value and units for regressions; a list of `{date, price, lower, upper}` for the forecast). Requests are validated by Pydantic schemas that mirror the notebook's predictor list, and regressions attach a warning when an input falls outside the training range, since a fitted line has no information beyond it. Endpoints, request bodies and example calls are listed in Appendix B.

3 Justification and explanation of the models

Each subsection says what the model is asked, why the technique fits, what had to be done to the data, and which of the models built in the notebook was kept and why. The numbers are the ones printed by the executed notebooks in the repository.

3.1 Bitcoin price: ARIMA

The question. What will bitcoin cost in the next few days?

Why ARIMA. There are no predictors here, only the history of the price, and ARIMA is the technique of the course for forecasting a series from its own past.

What the data needed. The price grew twenty-fold over the four years, and its daily jumps grew with it: a 10 % move was 20 dollars in 2015 and 250 in 2017. ARIMA needs a series with a stable mean and a stable spread, so two steps were taken. Taking the logarithm makes the spread stable, because it turns dollar changes into percentage changes. Taking the difference between one day and the next removes the trend. After those two steps the statistical test for stationarity passes.

What the model found. The daily percentage change of bitcoin has almost no memory: knowing today's change says almost nothing about tomorrow's. That is what economists call a random walk, and it means the best guess for tomorrow is roughly today's price. The notebook tried sixteen ARIMA orders and kept the one with the best AIC score, ARIMA(3,1,3), but none of its coefficients is statistically different from zero, so it is really a random walk with a few small extra terms.

What is kept. The ARIMA is kept because, unlike a plain "same as today" answer, it gives a range around the forecast that widens with the horizon. The assistant reads out the number and the range, and Section 4 shows honestly that the number itself is no better than repeating the last price.

3.2 Used car price: multiple linear regression

The question. How much does a used car sell for, given its showroom price, year, mileage, fuel, seller and transmission?

Why linear regression. A dealer wants to know how much each fact is worth, for example how much a year of age costs. A linear regression gives exactly that: one coefficient per fact, in units of price.

What the data needed. Most cars are cheap and a few are very expensive, so prices are lopsided. Fitting the price directly, the model gets the expensive cars badly wrong and its errors grow with the price. Fitting the logarithm of the price instead, with the logarithm of the showroom price and of the mileage, fixes that; the coefficients then read as percentages, which is the natural way to talk about depreciation.

What is kept. The logarithmic model, with an R^2 of 0.954 against 0.807 for the direct model and half the error. Cars outside the range seen in training (a much more expensive or much older car) are not predicted, because a straight line knows nothing beyond its data. The coefficient for CNG rests on two cars and is reported as unknown.

3.3 Telecom churn: neural network

The question. Will this customer cancel the service?

Why a neural network. A multilayer perceptron can combine the 30 facts about a customer (contract, services, bill, tenure) in ways a straight line cannot. The network has two hidden layers of 32 and 16 neurons and one output neuron that gives a probability.

Why ReLU. The activation of the hidden neurons is set explicitly. The notebook tries the four that scikit-learn offers (identity, logistic, tanh and ReLU) with the same architecture, on the training data alone, and they land within one standard deviation of each other: a cross-validated AUC between 0.830 and 0.839 with a fold moving it by about 0.014. The identity row matches the rest, which says the churn signal is largely linear, but a network with no non-linearity is a logistic regression and is not kept. Among the non-linear three ReLU is kept because its gradient does not fade on the positive side, and the other two buy nothing measurable here in exchange for their saturating sums.

What the data needed. Only about one customer in four leaves, so the network learns to say "stays" most of the time: at the usual cut of 0.5 it reaches 79 % accuracy but catches only half of the customers who leave. The notebook sweeps the threshold and shows that the ranking behind the probabilities is good (AUC 0.84): lowering the cut to 0.3 finds three leavers out of four at the price of twice the false alarms.

What is kept. The network at the 0.5 threshold, which is the most accurate cut, with the probability stored so the company can lower the threshold when a retention call is cheap next to a lost customer. The weak number is stated plainly: about four customers in ten on the list would not have left, so the list is a set of customers worth a call, not a verdict.

3.4 Wine quality: k nearest neighbours

The question. How good is a wine, from its laboratory sheet?

Why k-NN. It needs no equation: to score a new wine it looks up the k most similar wines in the training set and takes the most common score among them.

What the data needed. "Similar" means close in every measurement, but the measurements use very different scales (sulphur dioxide in the hundreds, density around 1). Without putting them on the same scale the distance is decided by sulphur dioxide alone, so every measurement was standardised. The second problem is the target: scores go from 3 to 9 but almost every wine is a 5, 6 or 7, and a vote among neighbours never picks the rare scores. Grouping the scores into low, medium and high gives every class enough wines to win a vote.

What is kept. The three-band model with $k = 39$. It gets the band right 59 % of the time, against 44 % for always answering "medium", and its mistakes are between neighbouring bands. The honest reading is that the laboratory sheet explains only part of what tasters score, so the assistant gives a band, not a number.

3.5 Flight delay: decision tree

The question. How late will this flight arrive?

Why a decision tree. A tree asks a chain of yes/no questions ("does it depart after 5 pm?", "is it this airline?") and predicts the average delay of the flights that answered the same way. Its rules can be read, so the assistant could explain why it expects a delay.

What the data needed. The file contains the departure delay of every flight. A tree that reads it predicts the arrival delay almost perfectly (R^2 of 0.95), but that is cheating: if you already know the plane left forty minutes late you do not need a model. The honest tree only reads what is known before leaving home: airline, route, month, weekday, scheduled hour and distance.

What is kept. The honest tree, with an R^2 of 0.05 and an average error of 36 minutes. The notebook says plainly that this is a weak model, and explains why: delays come from weather, breakdowns, late inbound aircraft and airport congestion, none of which is in the schedule. The assistant gives the expected delay together with that error.

3.6 Stroke risk: logistic regression

The question. Is this person at risk of a stroke?

Why logistic regression. It turns a set of facts (age, blood pressure, heart disease, glucose, body mass, smoking) into a probability between 0 and 1, and a threshold on that probability gives a yes or no.

What the data needed. Only 1 person in 20 in the file had a stroke. Trained as is, the model says "no" to almost everyone: 95 % accuracy, but it finds 2 strokes out of 62. For a screening tool that is useless. Telling the model that a missed stroke weighs as much as the nineteen healthy people it is outnumbered by (balanced class weights) fixes it.

What is kept. The balanced model. It finds 50 of the 62 strokes, at the cost of sending about 300 healthy people out of 1 216 to a check-up. Age matters five times more than any other fact. Because only about one positive in seven is a real stroke, the assistant presents a positive answer as a reason to see a doctor, never as a diagnosis.

3.7 Cirrhosis stage: support vector machine

The question. How advanced is this patient's liver disease, from blood tests and physical signs, without a biopsy?

Why an SVM. It draws the boundary between classes that leaves the widest gap on each side, and with the radial kernel that boundary can curve. Because it measures distances between patients, all the tests had to be put on the same scale first, and its two settings (C and γ) were chosen by cross-validation.

What the model found. Telling apart the four stages is not possible from these data: stages 2 and 3 have the same typical values on almost every test, and no setting of the model got above 45 % balanced accuracy. What the tests do show is whether the patient has reached cirrhosis (stage 4) or not.

What is kept. The two-class model, cirrhosis or earlier, right about three times in four. The four-stage model is saved next to it so the API can still answer with a stage, with its limitation attached. The data come from a clinical trial of the 1970s and 1980s, mostly women, so the model is a screen, not a diagnosis.

3.8 Avocado price: random forest

The question. What will an avocado cost in a given city, for a given type (organic or conventional) and week?

Why a random forest. A single decision tree memorises its training data: grown without limits it is perfect on the rows it saw and much worse on new ones. A forest grows hundreds of trees on random samples and averages them, which keeps what the trees learn and cancels what they memorise.

What the model found. The forest of 200 trees explains 90 % of the price with an average error of nine cents, and its out-of-bag score agrees. The type of avocado is by far the most important fact (organic costs about half a dollar more), followed by the quantities sold and the season.

What is kept. The forest, with one warning. Trained on 2015–2017 and asked about 2018, it explains only 41 % of the price, because a forest cannot guess a year it has never seen. So it is used to look up the expected price for a region, type and season, not to forecast next year.

3.9 Hepatitis C category: naive Bayes

The question. Does this blood panel look like a healthy donor, hepatitis, fibrosis or cirrhosis?

Why naive Bayes. It computes, for each category, how likely the observed lab values are, and picks the most likely category. It works well with few examples, which matters here: the file has 533 donors but only 24, 21 and 30 sick patients.

What the data needed. The Gaussian version assumes each lab value follows a bell curve within each category. Liver enzymes do not: most values are small and a few are huge. Taking the logarithm of every lab value makes them bell-shaped and raises the balanced accuracy from 0.65 to 0.71.

What is kept. The logarithmic model. It separates donors from patients almost perfectly and recognises cirrhosis four times out of five, but it confuses hepatitis and fibrosis with each other about half the time, so the assistant reads either of those as "early liver disease". Because the sick groups are so small, the numbers come from five-fold cross-validation rather than a single test set.

3.10 Body fat: multiple linear regression

The question. What is a person's body fat percentage, from measurements taken with a tape and a scale?

Why linear regression. The reference method (underwater weighing) is expensive; a formula with a few readable coefficients is what a trainer can actually use.

What the data needed. The file has thirteen measurements, but they mostly say the same thing: a bigger man has a bigger chest, waist, hip and thigh, all at once. When predictors repeat each other, a regression cannot decide how to share the effect between them and its coefficients become meaningless, some even with the wrong sign. The `Density` column was dropped because the target is computed from it. Two impossible rows (a man 75 centimetres tall, a body fat of exactly 0) were removed.

What is kept. A model with three measurements: abdomen, weight and wrist. It is as accurate as the model with all thirteen (average error of 3.9 points of fat against 3.8, on a test set too small to tell the difference) and its three coefficients can be read and trusted. The limitation is the error itself: a man predicted at 20 % may be at 16 or 24, enough to place him in a broad category but not to follow a diet week by week.

4 Analysis of results

4.1 Model metrics

Tables 3 to 7 gather the metrics of the model kept in each notebook, next to the literal model built first when the two differ. Every figure is computed on rows the model never saw: a 25 % random test split for most notebooks, a chronological hold-out for the two series, and five-fold cross-validation for hepatitis, whose sick classes are too small for a single split.

Forecast	Window	MAE (USD)	RMSE (USD)	MAPE (%)
ARIMA(3,1,3)	30 validation days	210.7	246.6	8.4
Naive (last price)	30 validation days	214.1	249.7	8.6
ARIMA(3,1,3)	7 unseen days, August 2017	236.1	282.1	7.6
Naive (last price)	7 unseen days, August 2017	233.2	284.9	7.5

Table 3: Bitcoin forecast against the naive forecast on the two hold-outs.

Model	Test rows	R^2	MAE	RMSE	Units
Car price, log model (kept)	72 cars in range	0.954	0.65	1.03	lakhs
Car price, raw price	72 cars in range	0.807	1.23	2.11	lakhs
Flight delay, schedule only (kept)	50 000 flights	0.050	36.0	56.6	minutes
Flight delay, with departure delay	50 000 flights	0.952	8.9	12.7	minutes
Avocado, random forest (kept)	4 563 random weeks	0.898	0.091	0.128	USD
Avocado, random forest	1 296 weeks of 2018	0.410	0.188	0.235	USD
Body fat, 3 predictors (kept)	59 men	0.583	3.87	4.64	% fat
Body fat, all 13 predictors	59 men	0.602	3.83	4.53	% fat

Table 4: Regression results. The avocado forest's out-of-bag R^2 is 0.893; the adjusted R^2 of the two body fat models is 0.561 and 0.487.

Model	Test rows	Acc.	Sens.	Spec.	Prec.	AUC
Churn, MLP (kept)	1 761 customers	0.790	0.529	0.885	0.624	0.840
Churn, MLP at threshold 0.3	1 761 customers	0.762	0.745	0.768	0.537	0.840
Stroke, balanced logistic (kept)	1 278 people	0.751	0.806	0.748	0.140	0.843
Stroke, plain logistic	1 278 people	0.952	0.032	0.999	0.667	0.845

Table 5: Binary classifiers at the 0.5 threshold, plus the churn network at 0.3 for reference.

Model	Question	Evaluated on	Acc.	Bal. acc.
Wine, k-NN, $k = 39$ (kept)	3 bands	1 333 test wines	0.589	0.547
Wine, k-NN, $k = 5$	7 scores	1 333 test wines	0.513	0.230
Cirrhosis, SVM (kept)	stage 4 or earlier	103 test patients	0.777	0.758
Cirrhosis, SVM tuned	stages 1 to 4	103 test patients	0.447	0.424
Hepatitis, Gaussian NB, log labs (kept)	4 categories	608 people, 5-fold CV	0.934	0.709
Hepatitis, Gaussian NB, raw labs	4 categories	608 people, 5-fold CV	0.914	0.653

Table 6: Multi-class classifiers.

Kept model	Recall by class
Wine, 3 bands	low 0.66, medium 0.63, high 0.35
Cirrhosis, stage 4 or earlier	earlier stages 0.82, cirrhosis 0.69
Hepatitis, 4 categories	donor 0.98, hepatitis 0.58, fibrosis 0.48, cirrhosis 0.80

Table 7: Per-class recall of the kept multi-class models.

4.2 Reading the numbers

Three models are strong. The car price model explains 95 % of the variance of the log price with an error of 0.65 lakhs, the avocado forest explains 90 % with an error of nine cents on a random split, and the hepatitis classifier tells a donor from a patient with a recall of 0.98. All three are recommended to the assistant without reservation inside the range of their training data.

Four models are useful screens. Churn (AUC 0.84), stroke (AUC 0.84), cirrhosis-or-not (balanced accuracy 0.76) and body fat (error of 3.9 points) separate their classes or place a person in a broad band, and each notebook states the precision or the error that makes its answer a reason to look closer rather than a verdict. In every one of them the decision that mattered was not the estimator but the treatment of the imbalance or of the multicollinearity: balanced weights turned a stroke model that found 2 of 62 strokes into one that finds 50, the threshold sweep shows the same lever for churn, and dropping ten collinear circumferences made the body fat slopes readable at no measurable cost.

Three models are honest negatives. The ARIMA does not beat the naive forecast on either hold-out; the difference is three dollars on a price of 2 500 and changes sign between the two windows, which is the result the flat autocorrelation announced: the daily return of bitcoin has no usable memory. The wine model misplaces four wines in ten because the laboratory sheet explains only part of a tasting score. The flight tree reaches an R^2 of 0.05 because the causes of delay are absent from the schedule. In all three the literal, accurate looking alternative (an ARIMA judged by its in-sample AIC, a tree that reads the departure delay) was built, shown, and rejected with the reason. The project treats reporting these outcomes as part of the scientific method rather than as failures to hide: the assistant tells the user the band around the bitcoin forecast, the band rather than the score for wine, and the 36-minute average error next to any delay.

Evaluation design matters as much as the model. The avocado forest drops from $R^2 = 0.90$ on a random split to 0.41 on the weeks of 2018, because a random split of a set of time series lets the model interpolate between neighbouring weeks. The hepatitis model is scored by cross-validation because its sick classes have 21 to 30 people. The car and body fat models exclude test rows outside the training range, and the API carries that rule into production as a warning on every out-of-range request.

4.3 Application and API

All ten models are served by the API and reachable from the assistant by voice and by text; the repository ships three spoken sentences per model that resolve in a single turn with Gemini and also with the local rules. The rule-based parser is covered by 82 unit tests (`app/intents/rules/tests`) that pass on the current revision; they check that every shipped prompt resolves locally without Gemini and that the extractors of the car price and stroke models read numbers, dates, Spanish number words and synonyms correctly.

The perception agents were validated by use rather than by a held-out set, since no labelled corpus of the team's faces or voices exists. Three observations from that use shaped the code. First, a USB webcam's first frames are nearly black while its exposure settles, so ten frames are discarded on open to avoid reporting "no face". Second, Whisper *small* in int8 runs at usable latency on a laptop CPU, and it mishears the wake word in a handful of stable ways, which are now in the wake-word table. Third, with seven emotion classes the winning probability of a webcam frame is

often below 0.5 and flips between neutral and a neighbour, so the emotion is sampled every two seconds rather than every frame, and only the winner drives the tone and the avatar.

4.4 Limitations and future work

The emotion classifier was reused from a previous course project and not retrained; its accuracy on FER-2013 was not re-measured here and its behaviour on the team's faces was only observed. A retraining on FER-2013 with a held-out evaluation, or a lighter architecture, would let the notebook of that model join the other ten. The flight and wine models would need data that is not in their files (weather, tasters). The bitcoin model would need exogenous predictors to beat a random walk, and the literature suggests those gains are small. Whisper *small* is a compromise between latency and accuracy on CPU; a GPU or the *medium* model would cut mishearings. Finally, the Gemini parser depends on a free-tier quota, which is why the local rules exist and why the model chain skips exhausted models for the rest of the session.

A Repository and reproduction

The repository is at `https://github.com/Jonathansl17/jarvis`. Its layout:

```
notebooks/   one folder per model, each with its executed CRISP-DM notebook
models/      one .joblib bundle per model, written by the notebooks, read by the API
             plus facial_emotion_cnn.onnx and haarcascade_frontalface_alt.xml
api/         FastAPI service, one adapter and one schema per model
app/         JarvisTEC desktop assistant (CustomTkinter, whisper, Piper, OpenCV)
docs/        this document (LaTeX)
```

Notebooks. Every notebook downloads its dataset with `kagglehub` or from a public URL, so it runs unchanged in Google Colab and locally, and its last cells write `models/<name>.joblib` and reload it for a sanity prediction.

```
python3 -m venv env && source env/bin/activate
pip install -r notebooks/requirements.txt
jupyter notebook notebooks/
```

API.

```
pip install -r api/requirements.txt
cd api && uvicorn app.main:app --host 0.0.0.0 --port 8000
```

Interactive documentation at `http://localhost:8000/docs`. Configuration through the environment variables `MODELS_DIR`, `HOST` and `PORT`.

Application.

```
pip install -r app/requirements.txt
cp app/.env.example app/.env     # optional GEMINI_API_KEY, API_BASE_URL
python -m app                    # with the API running
```

The first run downloads Whisper *small* (about 500 MB) and the Spanish Piper voice (about 60 MB). Other variables: `GEMINI_MODELS`, `MODELS_DIR`, `PIPER_CACHE_DIR`, `CAMERA_ENABLED`, `TTS_ENABLED`. The same steps work on Windows with `py -m venv env` and `env\Scripts\activate`.

Tests.

```
python -m unittest discover -s app -p "test_*.py" -t .
```

B REST API

Method	Path	Purpose
GET	`/health`	liveness, no model lookup
GET	`/models`	names of the loaded models
GET	`/models/{model_name}/schema`	JSON schema of that model's request body
POST	`/predict/{model_name}`	runs one model on a JSON body

Table 8: Endpoints of the API.

Optional fields (`null` or omitted) are filled with the value the notebook used for missing data. Classifiers answer with `label`, `probability` and the `threshold` applied; regressions answer with

the predicted value in the units of the notebook and `meta.warnings` when an input falls outside the training range; bitcoin answers with a list of `{date, price, lower, upper}`. Example calls:

```
curl -X POST localhost:8000/predict/car_price_regression \
  -H 'content-type: application/json' -d '{
  "present_price": 10, "year": 2015, "kms_driven": 40000, "owner": 0,
  "fuel_type": "Diesel", "seller_type": "Dealer", "transmission": "Manual"}'

curl -X POST localhost:8000/predict/body_fat_regression \
  -H 'content-type: application/json' -d '{"abdomen": 90, "weight": 175, "wrist": 18}'

curl -X POST localhost:8000/predict/stroke_classification \
  -H 'content-type: application/json' -d '{
  "gender": "Male", "age": 68, "hypertension": true, "heart_disease": false,
  "ever_married": "Yes", "work_type": "Private", "Residence_type": "Urban",
  "avg_glucose_level": 140, "bmi": 31, "smoking_status": "formerly smoked"}'

curl -X POST localhost:8000/predict/hepatitis_type_classification \
  -H 'content-type: application/json' -d '{
  "age": 45, "sex": "m", "ALB": 38, "ALP": 60, "ALT": 30, "AST": 90, "BIL": 30,
  "CHE": 5, "CHOL": 4, "CREA": 80, "GGT": 120, "PROT": 70}'

curl -X POST localhost:8000/predict/wine_quality_classification \
  -H 'content-type: application/json' -d '{
  "type": "white", "alcohol": 13.5, "density": 0.99, "free_sulfur_dioxide": 30,
  "total_sulfur_dioxide": 120, "volatile_acidity": 0.25, "residual_sugar": 2}'

curl -X POST localhost:8000/predict/flight_delay_regression \
  -H 'content-type: application/json' -d '{
  "month": 12, "day_of_month": 19, "day_of_week": 5, "dep_hour": 18,
  "carrier": "AA", "origin": "ORD", "dest": "LAX", "distance": 1745}'

curl -X POST localhost:8000/predict/bitcoin_price_regression \
  -H 'content-type: application/json' -d '{"days": 7}'
```

C Software stack

Component	Library and version
Notebooks	pandas 3.0.5, scikit-learn 1.9.1, statsmodels 0.15.0, matplotlib 3.11.1, seaborn 0.13.2, kagglehub 1.0.2, joblib 1.6.0
API	FastAPI 0.141.1, uvicorn 0.52.4, pydantic-settings 2.15.0
Application	CustomTkinter 6.0.0, faster-whisper 1.2.1, piper-tts 1.8.0, onnxruntime 1.30.0, opencv-python-headless 4.14.0, sounddevice 0.5.6, httpx 0.28.1, google-genai 2.23.0
Emotion model	VGG-style CNN, Keras weights exported to ONNX; Haar cascade `frontalface_alt` from OpenCV
Speech	Whisper *small*, int8, CTranslate2; Piper voice `es_MX-claude-high`
Intent parsing	Gemini (`gemini-3.6-flash`, then `gemini-flash-lite-latest`) with a rule-based fallback

Table 9: Versions pinned in the three `requirements.txt` files.